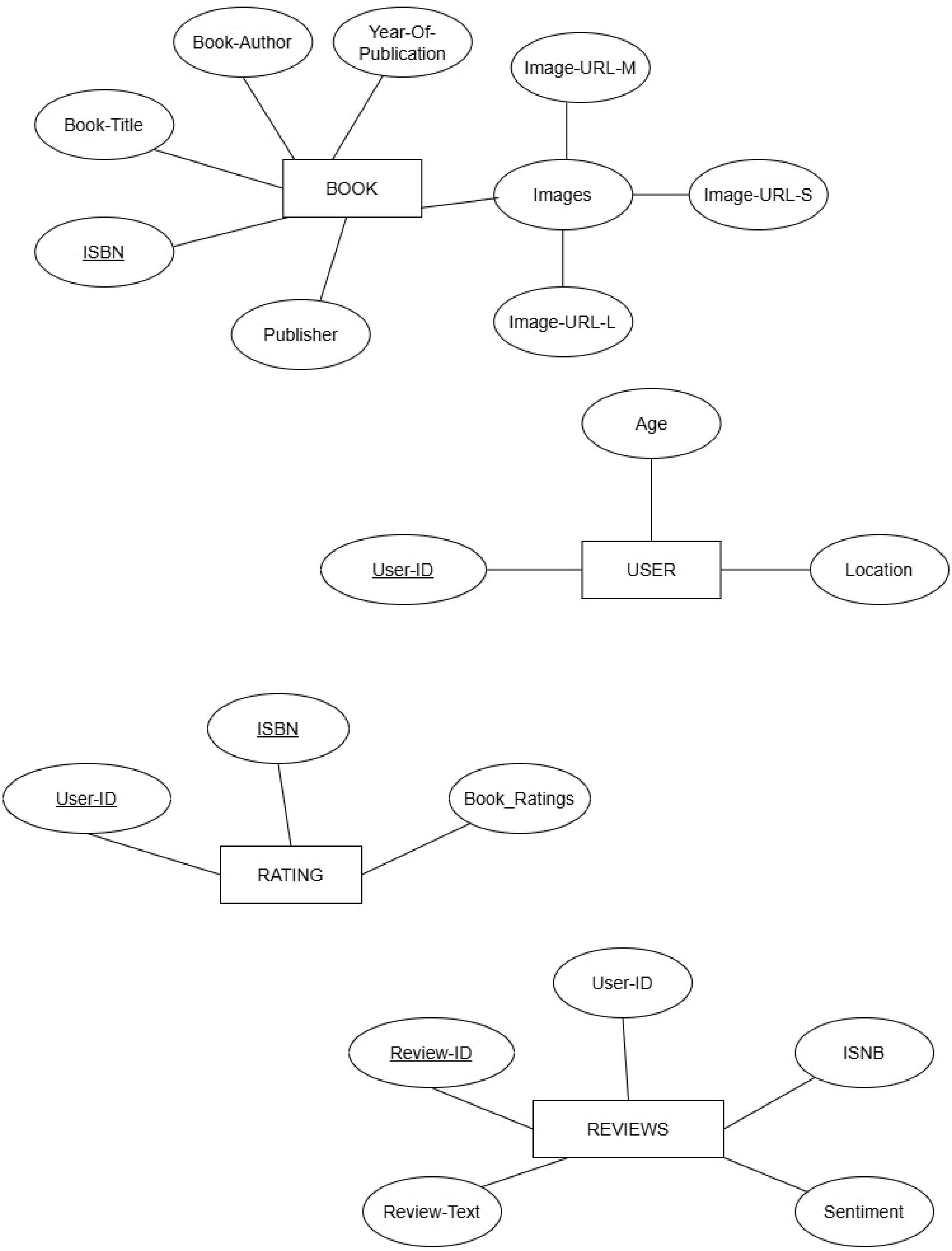
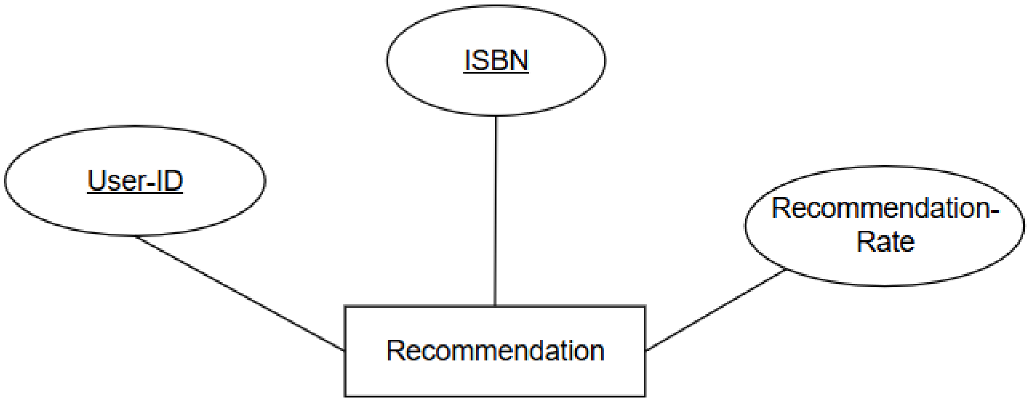
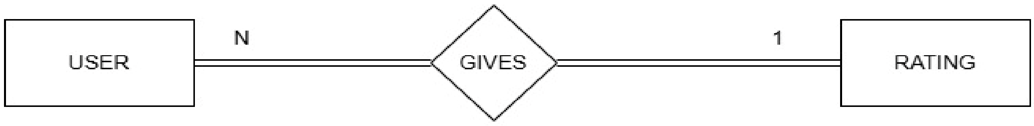
1. <u>ER Diagrams :</u>

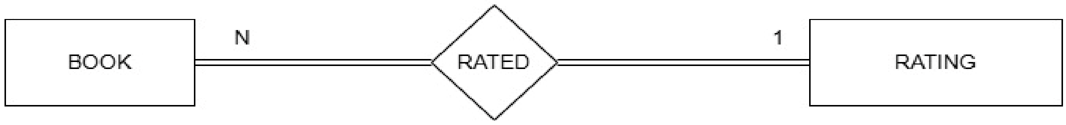
Entity types and Their Attributes

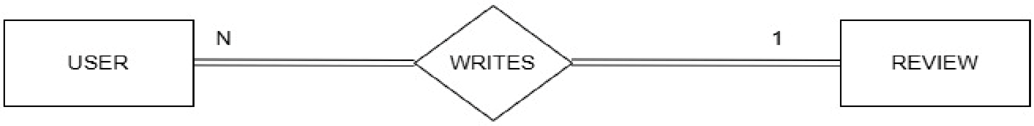


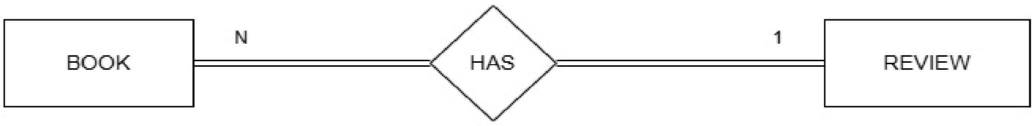


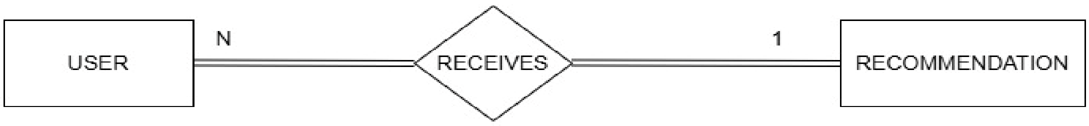
Relationship :



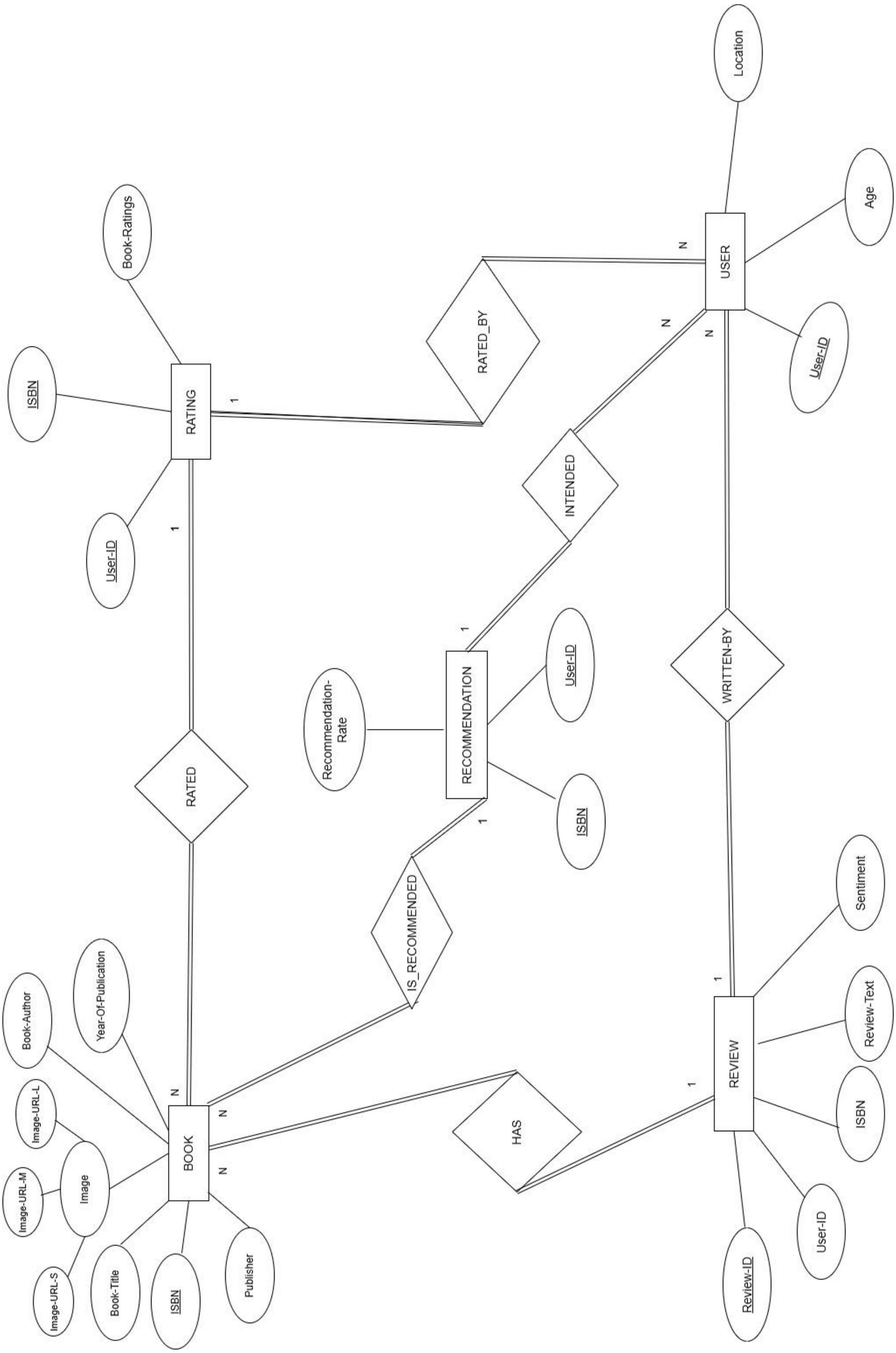




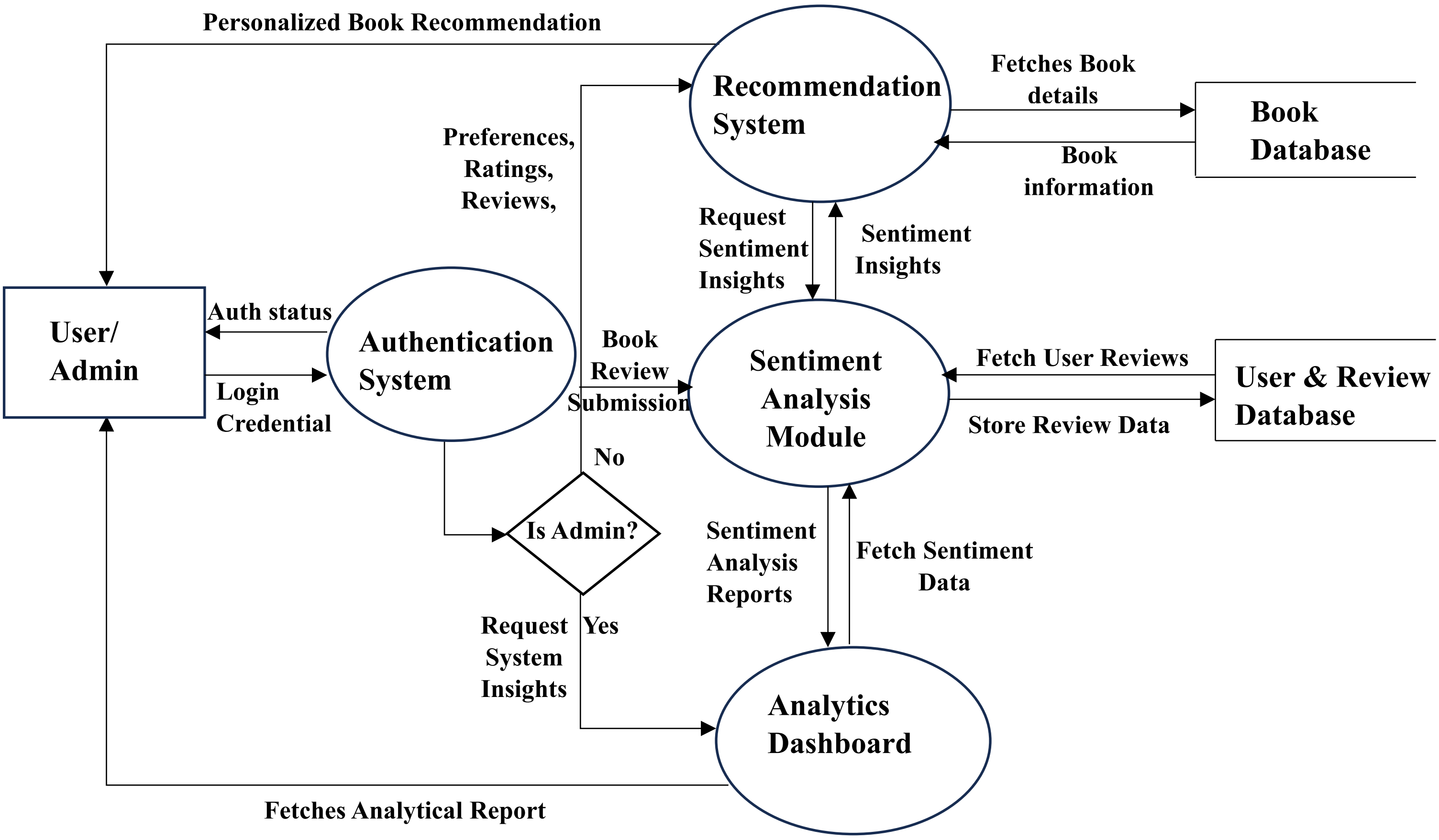




I. ER Diagram


USER
Location
Age
User-ID
RATED_BY
RATING
Book-Ratings
ISBN
User-ID
INTENDED
WRITTEN-BY
RECOMMENDATION
Recommendation-Rate
User-ID
ISBN
RATED
IS_RECOMMENDED
REVIEW
Sentiment
Review-Text
ISBN
User-ID
Review-ID
HAS
BOOK
Book-Author
Year-Of-Publication
Image-URL-L
Image-URL-M
Image
Image-URL-S
Book-Title
ISBN
Publisher

II. Data-Flow diagram :



2. <u>Dataset Analysis:</u>

I. Books dataset

- This dataset includes attributes like ISBN, Book-Title, Book-Author, Year-Of-Publication, Publisher, Image-URL-S, Image-URL-M, Image-URL-L.
- The dataset contains 271,360 rows and 8 Columns.
- ISBN attribute contains 271,360 unique values.
- Book-Title attribute contains 242135 unique values.
- Book-Author attribute contains 102023 unique values.
- Year-Of-Publication attribute contains 202 unique values.
- Publisher attribute contains 16807 unique values.

Distributions & Relationships:

- Book-Author and Publisher have high cardinality (many unique entries).
- Year-Of-Publication has a right-skewed distribution, with many values clustered between 1990–2005, and anomalies like 0 and 9999.
- Some authors and publishers are highly dominant.

II. Users dataset

- The dataset includes attributes like User-ID, Location, Age.
- The dataset contains 278,858 rows and 3 Columns.
- Statistical analysis of Age
  - Mean : 34.751434
  - Median : 32.0
- User Id attribute contains 278,858 unique values
- Age attribute contains 165 unique values.
- Location attribute contains 57,339 unique values.
- Most users are between 24 and 44 years old (IQR).

Distributions & Relationships:

- Age contains a wide range from 0 to 100+, but with noticeable outliers (e.g., 0, 1, 110).
- Majority of users fall between ages 20 to 40.
- Location is not standardized — includes city, state, and country in various formats.

III. Rating Dataset

- This dataset includes attributes like User-ID , ISBN, Book-Rating.
- The dataset contains 1,149,780 rows and 3 Columns.
- Statistical analysis of Book-Rating
  - Mean : 2.866950
  - Median : 0.0
- User Id attribute contains 105,283 unique values.
- Book -Rating contains 11 unique values.
- ISBN contains 340556 unique values.

Distributions & Relationships:

- Many Book-Rating values are 0.
- Ratings show a bimodal distribution: a spike at 0 and another at 8–10.
- Popular books tend to have more ratings .
- Few users rate a large number of books, while most users rate very few.

3. Data Issues:

I. Books dataset

a) Missing Values

- ISBN 0
- Book-Title 0
- Book-Author 1
- Year-Of-Publication 0
- Publisher 2
- Image-URL-S 0
- Image-URL-M 0
- Image-URL-L 3

b) Duplicate Values - 0

c) Inconsistencies in Year-Of-Publication datatype.

d) No outliers.

e) Anomalies :

- Invalid Publication Years: Found years like 0, 9999, or years outside realistic range (<1800 or >2025).
- Missing or Invalid Publisher Names: Some entries are blank or contain placeholders like "NaN" or "unknown".
- Suspicious Book Titles: Titles containing only punctuation (e.g., "...", "???") or very short ones.

II. Users dataset

a) Missing Values

- User-ID 0
- Location 0

- Age 110762

b) Duplicate Values - 0

c) No Inconsistencies.

d) Outliers in Age Column.

e) Anomalies :

- Unrealistic Age Values: Ages such as 0, 1, 122, 999 – outside the typical human age range.
- Inconsistent Locations: Entries like "n/a", ".", or malformed location strings.
- User-ID Anomalies: While unique, some IDs may not link properly with ratings or have repeated data.

III. Rating Dataset

a) Missing Values

- User-ID 0
- ISBN 0
- Book-Rating 0

b) Duplicate Values - 0

c) No Inconsistencies.

d) No outliers.

e) Anomalies :

- Ratings Outside Expected Range: Expected range: 0–10. Ratings of 0 are treated as implicit and should be filtered accordingly.
- Some users have rated the same book more than once — should keep the most recent or average.
- Ratings referencing User-IDs or ISBNs that don't exist in the Users or Books dataset.

4. Data Cleaning Requirements (Next Steps)

I. Books Dataset

- Remove invalid Year-Of-Publication entries like 0, 9999, or future years beyond 2025 and standardize.
- Standardize Book-Title (trim whitespace, fix casing, remove special characters).

- Clean Publisher and Book-Author fields by removing placeholders like "NaN", "Unknown".
- Check and fix ISBN format (ensure length is 10 or 13 characters and alphanumeric).
- Drop or correct rows with missing or inconsistent key fields (e.g., missing author, title).

II. Users Dataset

- Filter out unrealistic ages (e.g., below 10 or above 100).
- Handle missing or null Age values (either drop or impute with median).
- Clean and standardize Location
- Ensure unique User-IDs and consistent formatting.

III. Ratings Dataset

- Remove Book-Rating values of 0 if treating them as non-ratings or implicit feedback.
- Ensure all ratings fall between 1 and 10.
- Remove duplicate ratings for the same book by the same user (keep latest or average).
- Ensure all User-ID and ISBN entries match with those in Users and Books datasets.
- Drop rows with missing or inconsistent foreign key references.